

# Paper Review: Electricity and Gravitation—H. Weyl

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## 1 Bibliographical Information

WEYL, H. Electricity and Gravitation. Nature 106, 800–802 (1921).  
<https://doi.org/10.1038/106800a0>

## 2 Main Claim

Based on title, abstract, and the first few sentences, I think the main claim for this paper is:

A unification of gravity and electromagnetic forces. More precisely, absorbing electromagnetism into gravity, using the geometric structure of GR.

## 3 Why is this interesting?

Unification is a pursuit that follows the spirit of Maxwell, Lorentz, Einstein, Dirac, Glashow-Weinberg-Salam, Witten and so on. With the belief of nature is simple, beautiful, and harmonic. Specifically, I am interested in Weyl's early failed idea on the unification of E&M forces and gravity, which we still haven't succeeded in yet 100 years later. Why did Einstein disagree with Weyl's idea, and how the gauge field resurrected from QFT.

## 4 Guess how the result was obtained

Based on the title, and my main claim above: Since Gravity is acceleration, E&M forces should also be acceleration if we want to unify them. So Weyl might have introduced some "E&M mass" like gravitational mass, and let it be the same as inertial mass.

There are indeed two different understandings of the equivalence of inertial mass and gravitational mass, I prefer the latter:

**1,** Inertial mass is gravitational mass: then there is no inertia, it is just gravity. Any other force's effect on massive objects will have something to do with

gravity. However, this makes gravity too special that its status is superior to other fundamental forces. I don't think gravity will govern objects' motion caused by other forces, and I don't think gravity has a higher status than others.

2, Gravitational mass is inertial mass: then there is no gravity, it is just inertia. Then E&M mass is also inertial mass. And then follow the same path of GR, there might be a  $T_{E\&M}^{\mu\nu}$  as an analogy to  $T_G^{\mu\nu}$  in the gravitational field equation. And like GR, the geometry of some unknown thing (that Weyl had to introduce) manifests as the electromagnetism force. And there will arise a new field equation which indicates that the unknown thing's geometry on the left hand side tells the matter on the right hand side how to move, and the matter tells the unknown thing's geometry how to behave geometrically, as the famous Wheeler's saying goes.

## 5 My guess vs. actual paper

Unfortunately, this title and the first paragraph did not immediately yield the exact mechanism of the paper. Maybe because this paper is too old, and back to then, there was no standard structures like abstract and introduction. But anyways, Weyl failed to tell me what he wanted to show in this paper in the beginning. This paper first gave the motivation, and then explained the background concepts of "Guiding Field" and "affinely connected". Then moved into "metric" and claimed that when metrical connection was virtually varied, the simplest assumptions possible leads to Maxwell's Theory.<sup>1</sup> And Weyl also mentioned the connection of Maxwell's Theory and GR is by some simplest extra terms on "the magnitude of action". Weyl apparently didn't explain in detail and didn't cite his previous papers in which his theory was written. But then he argued Einstein's criticism on his previous works in the last part of this paper. The core is the "non-integrability" of the transfer distances in Weyl's theory. If this is true, 2 same rods coincide with each other will not coincide after they separated and met again, 2 same clocks at the same spacetime will not at the same time after they separated and met again. This is like the twin paradox. But that is explained by GR, this is not. Weyl's main argument is based on the example of magnetic needles in a magnetic field. The direction of each needle is not depending on the direction of it last moment. Using modern words, it is the spontaneous breaking of symmetry. And this process is Markovian. In comparison, the direction of the axis of a rotating top is non-Markovian. In my opinion this is because of continuous symmetry. Newton's laws are continuous symmetries. However, the magnetic needles case is discrete, breaking symmetry. Weyl didn't mention what I said above, but he used the needles' example to claim that if the rods and clocks adjust themselves at every moment, it is possible to solve the paradox.

Also, a good news is, in the second part of the paper, Weyl noted that the effects of gravitation also belong to the guiding field, as we should expect a

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<sup>1</sup>Weyl, Electricity and Gravitation, 1921

priori from our experience of the equality of gravitational and inertial mass.<sup>2</sup> While the guiding field is just inertia, his words matches with my results above in part 4, in the latter understanding of the equivalence of inertial mass and gravitational mass.

## 6 Regime of validity

The idea of non-integrability is an assumption Weyl had made. If he is right, the unification theory of gravity and Maxwell's theory will surprisingly arise. Based on this, I think we should pay attention to this theory. Even there is a paradox problem, the beauty itself may lead to deep structures. Does nature possess coincidence that after such a complicated processes everything is connected? I don't think so. If Weyl was wrong, I think it is just his understanding of the phenomenon his theory presents is wrong. If people have a new vision to re-understand the theory, the truth will emerge.

According to Weyl's enlightening explanation to the paradox, on the one hand, I think we should reexamine it from a modern perspective. I think in modern understanding of spontaneous symmetry breaking, his needle example can be re-interpreted. But it is possible that the new interpretation will lead Weyl's explanation to become valid or not. On the other hand, we should notice that Weyl lived in the era that Quantum Mechanics was not established.

Most importantly, Weyl's result contradicts experimental spectroscopic results. I don't think Weyl was trying to fix this fatal problem. He seems to be trying to explain that the failure of matching rods and clocks is possible, regardless of contradicting experiments. He just admitted it at the end of the paper. I think Weyl should try to find some mechanism to show that there is something happening to cause the experiments to not behave as he expected.

## 7 What to do next

As I mentioned above, we should find a new vision to understand this Weyl's beautiful theory. We should use a perspective with Quantum Mechanics and even more modern theories and ideas. Note that we still have not unified Gravity and electromagnetism forces yet! What Weyl did a few years later is to do the same thing on the phase of quantum fields. And a beautiful gauge theory emerges.

## 8 Mind palace entry

Weyl's theory does not match experiments, but we might review his example of spontaneous symmetry breaking, while the whole theory should be re-understood in a modern perspective.

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<sup>2</sup>Weyl, Electricity and Gravitation, 1921